

HealthConnect - Week 6 Project Summary

AnalystLab Africa Experience Lab | Data Science Track

Week 6 Goal

Week 6 focused on improving and validating the Week 5 no-show prediction model. I examined model errors, refined features, compared alternative models, and used findings from Data Analytics to support modelling decisions.

Key Analysis

Previous no-show history showed a clear pattern: the no-show rate increased from **46.3%** with no previous no-shows to **69.7%** with three. Appointments without reminders had a **54.6%** no-show rate compared with **49.9%** when reminders were sent. Diagnostic Tests showed the clearest reminder-related difference.

Error Analysis

The Week 5 Logistic Regression performed better for patients with clearer risk patterns. Booking lead time and previous no-show history helped explain many predictions, while less obvious no-shows were harder to identify.

Model Improvement

I tested targeted interaction features and compared the baseline Logistic Regression, refined Logistic Regression, Random Forest, and Gradient Boosting. Gradient Boosting gave the strongest overall classification results.

Candidate Model Results

Metric	Gradient Boosting
Accuracy	65%
No-Show Recall	65%
No-Show F1-score	65%
ROC-AUC	0.68

Threshold Validation

At the default threshold of **0.50**, the model provided balanced performance. Lowering the threshold to **0.45** increased no-show recall to about **74%**, but reduced precision and accuracy. This may be useful if HealthConnect prioritizes identifying more potential no-shows.

Cross-Track Integration

Data Analytics findings on booking lead time, previous no-show history, reminders, appointment type, and distance helped guide deeper analysis and feature decisions. The final candidate model and modelling requirements were also prepared for ML Engineering handoff.

Main Challenge

The improvement was modest. The strongest models remained around **0.68 ROC-AUC**, suggesting that the available features provide useful but limited predictive information.

Week 7 Focus

Week 7 should test model stability, the proposed decision threshold, false negatives, pipeline compatibility, reproducibility, and suitability for HealthConnect administrative use.

Week 6 Status: Model improved, compared, validated, and prepared for the next testing stage.